

Mack Thompson

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EDUCATION

Auburn University

May 2027

B.S. in Computer Science & Mathematics, Minor in Business

GPA: 3.76/4.00

- Competitive Programming Team, Association for Computing Machinery (ACM), TFAS Charity Boxer, ΔXA fraternity

EXPERIENCE

Research Fellowship (2x)

Feb 2025 – Present

Auburn University – Computer Vision & Machine Learning

Auburn, AL

- Developing sequence-based ML model on EV battery telemetry to forecast temperature evolution and identify stress.
- Scraped dermatology dataset of 5k+ labeled images, applying augmentation and ETL preprocessing to stabilize training.
- Uncovered misclassification biases linked to skintone after evaluating EfficientNet-B0 with grouped confusion matrices.
- Led development of a CBT-based mobile therapy app end-to-end as one of two engineers, integrating HCI principles.

Software Engineer Intern

Sep 2025 – Present

OCV Apps – Public safety mobile apps

Opelika, AL

- Designed a centralized factory system to replace 500K+ lines of duplicated manifest JSON across 30+ apps, eliminating repetitive updates and reducing content maintenance effort by an estimated 80%.
- Built analytics dashboards aggregating 250K+ monthly events across 30+ production apps revealing low engagement in the wellness apps, driving the decision to integrate into higher-traffic platforms, reducing onboarding effort by 50%.

Software Engineer Intern

May 2025 – Aug 2025

Kahua Inc. – SAAS for construction projects

Alpharetta, GA

- Decreased automated test development time by 30% by enabling code modification without resetting the driver.
- Improved deployment reliability by expanding an Azure DevOps testing pipeline, automating 20+ key client workflows.

Full-Stack Web Developer

Jul 2025 – Sep 2025

Jup Jewels – e-commerce

Opelika, AL

- Increased site traffic by 25% after redesigning the storefront using customer feedback & owner collaboration.
- Created a custom jewelry-builder that became the best selling product & generated \$4K revenue in 30 days.

Founder

Jun 2021 – Dec 2024

Sports Camp

Atlanta, GA

- Scaled to 100+ campers & 6 staff each summer, generating \$100K+ in profit before transferring ownership.
- Deployed customer response agent with AWS Pub/Sub, automating 75% of client inquiries & eliminating wait periods.

PROJECTS

Geopolitical Trend Forecaster

Oct 2025

Python, PyTorch, Pandas, Streamlit

[View Repository](#)

- Built ETL pipeline from 200+ years of economic, demographic, military, & trade datasets from open-source archives.
- Developed quantitative scoring algorithms for country strength & geographical advantage to model competitiveness.
- Trained deep temporal models (TCNs) to project geopolitical shifts & relative country standings 30 years into the future.

Social Scheduling Platform

Aug 2024

React, postgresSQL, WebSockets, AWS, Docker

[View Repository](#)

- Built a calendar-sharing & messaging platform with a WebSocket server for real-time updates with a React frontend.
- Designed a modular RESTful API with PostgreSQL, supporting CRUD operations for users, events, and messaging.
- Containerized the full stack with Docker and deployed it on AWS EC2 using environment-based configuration and Docker Compose orchestration.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C/C++, C#, HTML/CSS

Frameworks & Libraries: React, Flutter, PyTorch, Pandas, Selenium, NumPy

Tools: AWS, Git, Azure, Node, PowerBI, SQL, Bash, REST, Firebase, Docker, PostgreSQL, SQLite, Excel

Awards: ICPC Southeast regional 3rd Place 2023 & 2025, GHSA AAA Regional 153 lbs Wrestling Champion 2023